

Software Quality Factors Analysis Proposal

The application I will be testing is a C++ desktop app, created with Qt. It's a Reports Manager for a company. It has a variety of sorting and filtering options so that the client can easily generate reports with the data introduced in the app.

The Quality Model I've chosen is the **ISO 9126** from 2001

Functionality	Reliability	Usability	Efficiency	Maintainab.	Portability
Suitability	Maturity	Understand	Time behav.	Analysability	Adaptability
Accurateness	Fault tolerance	Learnability	Resource beh.	Changeability	Instability
Interoperability	Recoverability	Operability		Stability	Conformance
Compliance				Testability	Replaceability
Security					

The factors I've chosen are:

- **Usability** - the degree to which a software is easy to use.
- **Maintainability** - the ease with which the software can be modified.
- **Portability** - the ease with which a software can be migrated from one environment to the other.
- **Functionality** - degree to which software satisfies stated needs.

This is how I plan to measure them:

Usability - is of course important for any application. For my project Usability with Understandability and Operability is especially important as the app will be used by users with relatively low software experience, so it must be easy for them to learn to use it.

This is how I will measure Usability:

- I will define some usability criteria such as "X out of Y users must be able to perform task Z in a certain amount of time or with a certain number of errors (errors resulted from a lack of system understanding and not software errors)(like clicking the share icon instead of copy because you don't understand what the symbols mean)"
- Then I will have a few users that match the target user type for the application, test it.

- I will observe how they react while they use the app (if they feel frustrated or not).
 - And then I will interview them about the app, making sure they know I am evaluating the app and NOT their performance, so that they feel comfortable talking freely about the app's performance.
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Portability - the app might be used on different operating systems, specifically different versions of Windows, so it must perform well regardless of environment.

This is how I will measure Portability:

- I will test if all the app's functionalities work on Windows, MacOS and Linux, and if it looks the same (and especially on different versions on Windows).
 - Measure exactly how long it takes to deploy and install the application on all 3 OSs.
 - Check if there are any differences if using different cpp compilers (GCC, MSVC)
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Maintainability - I will have to maintain this application for quite a while, and make sure it continues to run smoothly, so any bugs that might appear (maybe due to external factors) are easy to fix (meaning the architecture is well implemented)

This is how I will measure Maintainability:

- I will measure Cyclomatic complexity, Inheritance Tree Depth, Coupling using CPPMetrics plugin for VisualStudio or Sourcemeter (whichever plugin works).
 - I'll try to measure technical debt as well, if SonarQube works (might have to use a trial of the paid version).
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Functionality - I have already struggled a bit while understanding the requirements of this application. I thought the client wanted something else initially and I had to make lots of changes as I was going. So measuring functionality would be important to see if in the end I've managed to encapsulate what the user wanted.

This is how I will measure Functionality:

- Measuring how many of the original requirements have been met
- Check how many defects (bugs) the app has per functionality? or per X amount of lines of code?